

MA40001

Philip Murray

Week 7 - Presentations

Writing a presentation in Quarto

- ▶ Fork the MA40001Resources repository
- ▶ Copy the template presentation into a different folder
- ▶ Render the presentation
- ▶ Edit the slides and render

What happens in Week 11?

- ▶ You give an assessed presentation to 3 or 4 staff members + other students in your presentation session
- ▶ 9 minute presentation + questions
- ▶ Assessed on:
 - ▶ Mathematical content
 - ▶ Presentation style

What makes a good presentation?

- ▶ Know your audience
- ▶ Strong narrative
- ▶ Speak clearly and enthusiastically
- ▶ Prepare but do not over prepare
- ▶ Present the *correct amount* of material
- ▶ Engage with the audience (project to the back of the room, make eye contact, show enthusiasm for the work!)

Narrative

Suggestion:

- ▶ Present an overview (Background)
- ▶ Hone in on a problem (Aims)
- ▶ Solve the problem (Actual work)
- ▶ Place results in context (Discussion - Reflect on Aims, Future work)

What makes a good slide?

- ▶ Succinct title
- ▶ Snappy statements
- ▶ Diagrams/figures preferable to text
- ▶ Conclusion
- ▶ Rule of thumb: 1 minute per slide

What should I say?

- ▶ Use bullet points as memory aid for the main points you need to cover in the slide
- ▶ Figures/diagrams .. say what you see - talk the audience through figures ... “In the figure y is plotted against t . We can see the following behaviour: ...”

Questions?

Be prepared to defend content that you put up.

Possible questions:

- ▶ Place the work in context
- ▶ Details of a technique used
- ▶ Particular features of a figure
- ▶ Explain mathematical detail (e.g. what does the second term in equation 2 represent)

Week 6 - Python

Previously

- ▶ Using Python in Quarto
- ▶ Defining variables
- ▶ Defining containers (sets, lists, arrays, dictionaries)
- ▶ Writing logical statements `q`
- ▶ linear algebra (numpy)
- ▶ Plotting (matplotlib)

This week

- ▶ Using Python libraries
 - ▶ Symbolic calculations (sympy)
 - ▶ Finding the local minimum of a function of one variable (scipy)
 - ▶ Solving systems of ODEs (scipy.integrate)
 - ▶ data analysis (pandas)

Week 5

Recap

1. Quarto

- ▶ Using Quarto to render .qmd file into .html and .pdf formats
- ▶ Add document elements (sections, figures, tables, cross references, citations, schematic diagrams etc.)

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- ▶ Use Python libraries

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 - ▶ Using Quarto to render .qmd file into .html and .pdf formats
 - ▶ Add document elements (sections, figures, tables, cross references, citations, schematic diagrams etc.)
2. Python
 - ▶ Integrate Python code in a Quarto document
 - ▶ Use Python libraries
3. Github
 - ▶ Create and clone repositories
 - ▶ Fetch, pull, commit and push
 - ▶ Fork repositories
4. Install VSCode/Python/Github on your own computer/H1 desktop.

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1. Strip out figures/table/appendices. Does the writing stand alone,i.e. is there a coherent narrative to the presented work?

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4. Equations require punctuation.
5. What makes a good figure caption?
6. Mathematical functions: $\sin(x)$ rather than $\sin(x)$.

Tasks for this week

1. *Fork* blog and final report templates from github
2. Using the terminal in VSCode to render blog and final report (see bottom of Chapter 3).
3. Use Python libraries (e.g. Pandas, Matplotlib, numpy, sympy, scipy).

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- ▶ copying another's work;
- ▶ the use of another's ideas without acknowledgement.

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- ▶ code that you have not written must be referenced.

Generative AI

Generative AI uses statistical patterns in data upon which it has been trained to provide responses to natural language queries.

You might use it:

- ▶ to generate graphics/visual content
- ▶ to summarise long texts to check your understanding
- ▶ as a debating partner
- ▶ to generate questions from an article/text to check your understanding
- ▶ for debugging code
- ▶ for improving grammar

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1. We want to develop critical thinking and problem solving skills!
2. You are expected to defend the choices/methods/techniques that you use.
3. Generative AI can sound convincing but be factually incorrect (e.g. references)
4. Using generative AI to generate content without attributing the source is a form of plagiarism.

AI statement (University policy)

1. Fill in an AI statement in report appendix
2. Provide a record of
 - ▶ prompts
 - ▶ responses
 - ▶ how you used the output.
3. Be able to defend the work that you present (presentation/poster/viva).

Week 4

Tasks

1. Install software on H1 pc (see Chapter 1)
2. Learn to use github for file management (see Chapter 6 in notes)
3. Continue with Python exercises (Chapter 5)
4. Learn how to write a blog using the template
5. Submit a pdf: a quarto write up of your work so far in Python.

Week 3

Tasks

General:

- ▶ If you do not already have one, sign up for an account on <https://github.com>. We will use this next week.
- ▶ Install VSCode on your personal computer. Then install the Quarto extension on VSCode.
- ▶ Fill in feedback form linked on MyDundee announcement.

Python:

- ▶ Start embedding Python codes in your Quarto document
- ▶ Work through the exercises up to for loops and if statements.

Week 2

Tasks

Use Quarto to generate a pdf with:

- ▶ title, author
- ▶ section headings
- ▶ figures
- ▶ tables
- ▶ equations using latex
- ▶ schematic diagrams
- ▶ cross referencing of different objects
- ▶ bibliography
- ▶ appendices

Project narrative (suggestion)

- ▶ One sentence summary of your project
- ▶ Why is the topic important?
- ▶ What is the background to your project (place the work in historical context)?
- ▶ What are the project Aims?
- ▶ What work did you do to address the Aims?
- ▶ Outlook, reflection (recap the project, were the aims addressed?, what are the limitations of the work? what would come next?)

Disclaimer: there are other ways to structure a project narrative.
Discuss this with your project supervisor.

Weeks 1 and 2

Tasks

- ▶ Get assigned your project
- ▶ Set up a meeting with your project supervisor
- ▶ Devise a plan for how you are going to work on your project
- ▶ Read Chapter 1 of lecture notes
- ▶ Start using Quarto (Chapter 2 of lecture notes)

Modules Aims:

- ▶ investigate a new mathematical topic
- ▶ develop communication skills
- ▶ develop independent study skills
- ▶ critical analysis
- ▶ develop programming skills

Module Structure

Your project content

- ▶ work independently
- ▶ meet with your project supervisor (you need to discuss and arrange a meeting schedule).

Writing/presentation

- ▶ weekly sessions developing communication/writing/programming skills (see timetable).

Optional

- ▶ 2 p.m. on Monday is Maths seminar - you are encouraged to attend but it is optional.

Module assessment

- ▶ presentation (10%)
- ▶ interim report (15%)
- ▶ poster (10%)
- ▶ final report + *viva* (65%)

Template projects and presentations will be available.

How did projects get assigned?

- ▶ Algorithm maximises overall satisfaction (tries to get as many students as high up preference list as possible) and is constrained by supervisor workload
- ▶ If you are not satisfied, please come and discuss with me.

Feedback

- ▶ Formative assessment. You will receive feedback
 - ▶ on submitted weekly work
 - ▶ practice presentation sessions
 - ▶ from your supervisor in meetings
 - ▶ documents that your supervisor reads in advance of submission
- ▶ Summative assessment
 - ▶ You will receive an email with grades and feedback from examiners

Any potential issues

- ▶ It's a 30 credit module
- ▶ Speak with your project supervisor
- ▶ Come and speak with me
- ▶ SSLC